BTW Sandpile

Abelian structure & the identity configuration

Why the order of topplings does not matter — and what the neutral element looks like

Dhar (1990) · Levine & Peres (2010) · **Levine & Propp (2010)**

The abelian property — Dhar's theorem (1990)

Theorem (Dhar 1990)

Let σ be any configuration and let grains be added at sites $i_1, i_2, \ldots, i_n$ (with repetition allowed). After full relaxation, the final stable configuration does not depend on the order of topplings.

$$\sigma + e_{i_1} + e_{i_2} + \ldots + e_{i_n} = \sigma + e_{i_{\pi(1)}} + \ldots + e_{i_{\pi(n)}}$$

for any permutation π of $\{1, \ldots, n\}$

Consequence

The set of stable configurations forms a commutative monoid under grain-wise addition + relaxation.

Physical intuition

Local rule

each toppling only moves grains to neighbours

Commuting topplings

two non-adjacent topplings are independent and commute trivially

Adjacent sites

each site topples the same total number of times regardless of order

Dhar, Phys. Rev. Lett. 64, 1613 (1990)

The sandpile group $\mathcal{G}(\Lambda)$

Elements

The group elements are the recurrent configurations $\mathcal{R}$ on the lattice Λ with a fixed sink node at the boundary.

$$|\mathcal{G}| = |\mathcal{R}| = \det \Delta$$

Group law

For $a, b \in \mathcal{R}$:

$$a \oplus b := (a + b)^\circ$$

Add grain-wise, then relax to the stable recurrent configuration.

Group axioms

Closure: $(a+b)^\circ \in \mathcal{R}$

Associativity: from abelian property

Commutativity: from abelian property

Identity: $\exists!\, e \in \mathcal{R}$ (see next slides)

Inverse: $\exists\, a^{-1} \in \mathcal{R}$ for every a

The group order $|\mathcal{G}| = \det \Delta$ grows exponentially with the grid size N — e.g. for a 4×4 grid $|\mathcal{G}| = 4{,}096{,}000$.

Dhar (1990) · Björner, Lovász & Shor (1991) · Levine & Peres (2010)

Computing the identity — algorithm

Algorithm (Creutz 1991)

1. Start from the all-2 configuration $\sigma_max = 2 \cdot \mathbf{1}$
2. Add σ_max to itself, relax: $\sigma_1 = (\sigma_max + \sigma_max)^\circ$
3. Add σ_max again, relax: $\sigma_2 = (\sigma_1 + \sigma_max)^\circ$
4. Repeat until σ_k = σ_{k+1} (fixed point)
5. The fixed point is the identity e

Why it works

The all-2 configuration is the maximum stable element (all sites at z_crit − 1 = 3 for 2D BTW).

Adding it to itself produces a fixed point because:

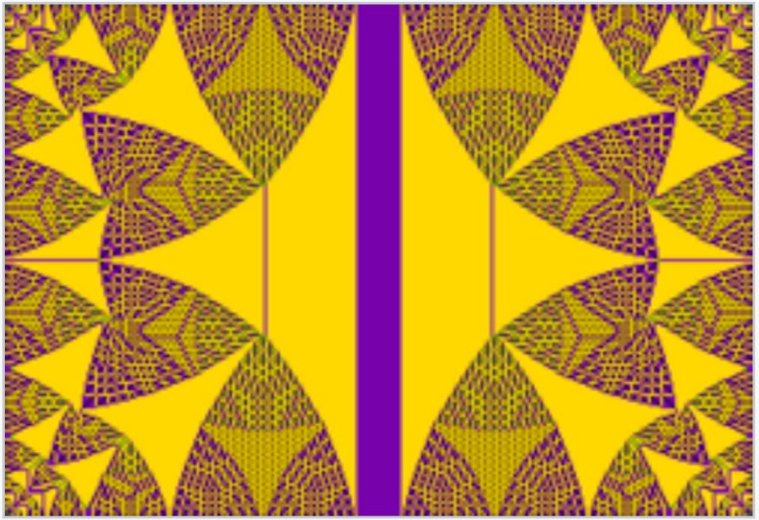
$$e \oplus \sigma_max = \sigma_max$$

so the fixed point must satisfy the identity axiom $e \oplus a = a$.

Complexity: $O(N^4)$ topplings for an N×N grid

Creutz (1991) · Levine & Peres (2010) · Also: Caracciolo et al. (2008)

The identity configuration



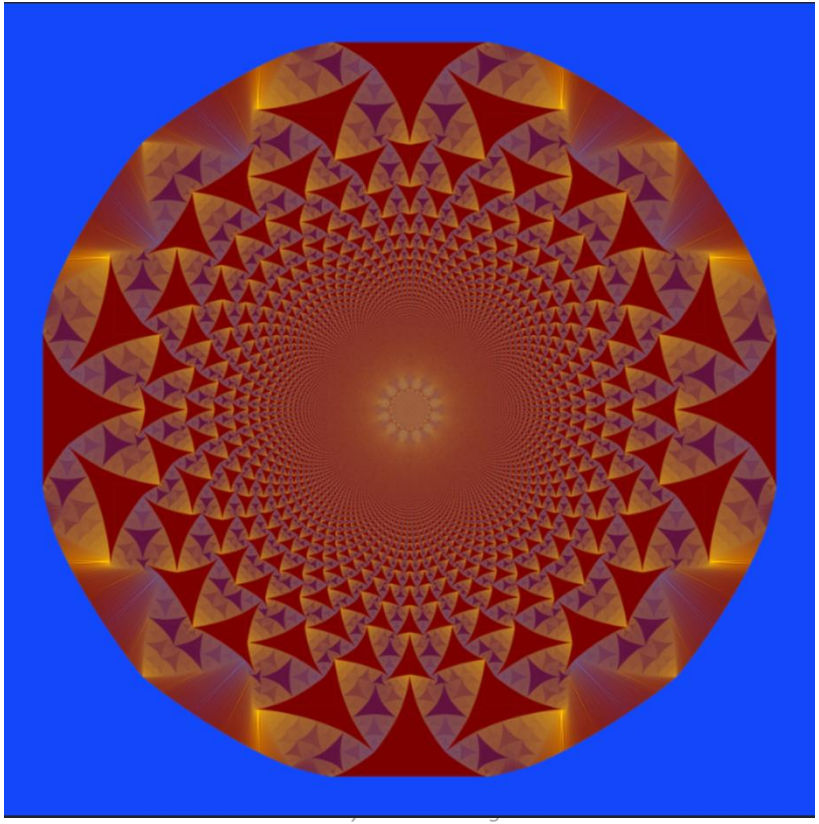
The identity element of the sandpile group of a rectangular grid. Yellow pixels correspond to vertices carrying three particles, lilac to two particles, green to one, and black to zero.

https://eavan.blog/posts/big-identity-sandpiles.html?utm_source=hnblogs.substack.com

https://higgsmass.medium.com/the-science-of-complexity-9940e0f902c1

https://www.pawelmandera.com/2017/12/24/sandpile/

Fractal structure of the identity



Self-similarity (Levine & Peres 2010)

The identity on a grid of size 2N is related to the identity on a grid of size N by a precise scaling + superposition rule.

Fractal dimension

The boundary between regions of constant height has fractal dimension:

$$d_f \approx 1.21 \pm 0.01$$

(Ostojic 2003, numerical estimate)

Summary — from abelian property to fractal identity

#	Concept	Result	Reference
1	Abelian property	Final state independent of toppling order	*Dhar (1990)*
2	Sandpile group $\mathcal{G}(\Lambda)$	Recurrent configs form a finite abelian group	*Dhar (1990), Björner et al. (1991)*
3	Group order	$\lvert\mathcal{G}\rvert = \det \Delta$ (matrix-tree theorem)	*Kirchhoff; Dhar (1990)*
4	Identity algorithm	Fixed point of $\sigma \mapsto (\sigma + \sigma_max)^{\circ}$	*Creutz (1991)*
5	Fractal identity	Self-similar pattern; $d_f \approx 1.21$	*Ostojic (2003); Levine & Peres (2010)*
6	Scaling limit (PDE)	Odometer → obstacle PDE $\Delta u = -2 \cdot \mathbf{1}$	*Levine & Peres (2010); Pegden & Smart (2013)*